

INSIGHTS

“What’s in a name?” Clarifying the identity of ROR γ ⁺ antigen-presenting cells

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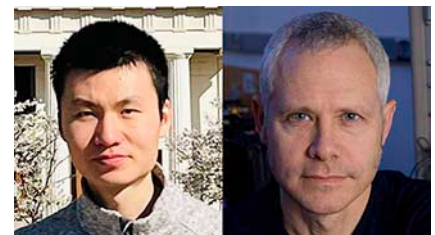
Recent publications have demonstrated that antigen-presenting cells (APCs) targeted by *Rorc*-cre (also known as ROR γ -cre) are required for the induction of peripheral regulatory T (pTreg) cells in response to commensal and dietary antigens. In this issue of *JEM*, Sun et al. (<https://doi.org/10.1084/jem.20250573>) provide key insights into the identity of these cells, revealing that *Rorc*-cre-traced APCs include group 3 innate lymphoid cells (ILC3s), dendritic cells (DCs), and extrathymic AIRE-expressing cells (eTACs). Their work highlights eTACs as critical for inducing ROR γ ⁺ pTregs specific to food antigens, while implicating DCs in the generation of ROR γ ⁺ pTregs.

A study by Sun et al. (2025) in this issue of *JEM* adds new insights into the rapidly emerging field of tolerogenic APCs, a journey the Gardner group began in 2008 (Gardner et al., 2008, 2013; Wang et al., 2021). A seemingly separate thread arose in 2013 with a report that ILC3s could mediate tolerance to commensal organisms (Hepworth et al., 2013, 2015). More recently, various threads were merged when three copublished studies concluded that APCs expressing retinoic acid receptor-related orphan receptor- γ t (ROR γ t) are indispensable for peripheral regulatory T (pTreg) cell differentiation against commensal antigens (Akagbosu et al., 2022; Kedmi et al., 2022; Lyu et al., 2022). Specifically, *Rorc*-cre-driven deletion of MHC class II (MHCII), chemokine receptor CCR7, or integrin α v β 8 resulted in loss of ROR γ ⁺ pTregs. These studies primarily focused on *Rorc*-expressing cells to identify these APCs (Akagbosu et al., 2022; Lyu et al., 2022). However, it is crucial to recognize that *Rorc*-cre targets not only cells currently expressing *Rorc*, but also *Rorc*-negative cells with past *Rorc* expression. As such, these strategies may have overlooked key APC populations.

To address this limitation, Sun et al. took an unbiased approach and profiled *Rorc*-cre-traced APCs by single-cell sequencing.

Their analysis revealed three principal populations: ILC3s, DCs, and eTACs (Sun et al., 2025). Notably, this approach captured a substantial population of DCs that do not express *Rorc* at the time of profiling, consistent with a recent report that DCs with *Rorc* expression history outnumbered those actively expressing *Rorc*(γ t)-GFP reporter (Narasimhan et al., 2025). Functional studies by Sun et al. demonstrated that MHCII presentation by *Rorc*-cre-targeted cells is required for pTreg induction in response to food antigens. This requirement was not attributed to ILC3s, as MHCII^{AROR γ t} + *Il7*^{-/-} mixed bone marrow chimeras (BMCs) lacking MHCII⁺ ILC3s displayed intact pTreg differentiation. In contrast, diphtheria toxin-treated MHCII^{AROR γ t} + AireDTR mixed BMCs, which lack MHCII⁺ eTACs, showed partial impairment of pTreg differentiation (Sun et al., 2025). While the requirement for the AIRE protein is still to be determined, the current study by Sun et al. suggests that eTACs are a key mediator of oral tolerance.

Collectively, these findings affirm that antigen presentation by some *Rorc*-cre-targeted APCs is necessary for pTreg differentiation in the contexts of commensal (Akagbosu et al., 2022; Kedmi et al., 2022; Lyu et al., 2022; Fu et al., 2025), and dietary and oral tolerance (Parisotto et al., 2024,



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Preprint; Fu et al., 2025; Rodrigues et al., 2025; Sun et al., 2025). Results from MHCII-ON^{ROR γ t} mice suggest that these APCs are not only necessary but also sufficient for pTreg induction in response to microbial antigens (Kedmi et al., 2022; Lyu et al., 2022). In addition to MHCII, other surface proteins such as CCR7 and α v β 8 are also required. One study showed that CCR7 on *Rorc*-cre-targeted cells is necessary for pTreg induction to a commensal organism (Kedmi et al., 2022). However, migration of CCR7^{AROR γ t} Janus cells (JCs) to mesenteric lymph nodes appeared normal (Kedmi et al., 2022). With improved knowledge about these APCs, it will be interesting to revisit whether CCR7 deficiency impairs the migration of other *Rorc*-cre-traced APCs. α v β 8 is known to activate transforming growth factor β (TGF- β), which is essential for pTreg differentiation (Travis et al., 2007).

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intact pTreg generation (Russler-Germain et al., 2021). This suggests that IRF8-dependent, *Zbtb46*-cre-targeted APCs other than cDC1s may be responsible for pTreg induction. It will be interesting to determine whether ROR γ ⁺ PRDM16⁺ APCs rely on IRF8 for their development or function. Second, beyond antigen presentation and $\alpha\text{v}\beta$ 8-mediated TGF- β activation, what other mechanisms endow this rare population of APCs with their robust tolerogenic capacity? Addressing this question may help harness these cells for therapeutic purposes. Third, do these cells function more broadly in tolerance beyond pTreg induction? For example, earlier studies on ROR γ ⁺ APCs suggested that ILC3s can facilitate tolerance via negative selection and IL-2 sequestration (Hepworth et al., 2013, 2015). These and other aspects of tolerance warrant revisiting. Finally, ROR γ ⁺ PRDM16⁺ APCs have been identified in various organs such as the spleen,

lung, and tonsil (Ulezko Antonova et al., 2023; Narasimhan et al., 2025). Their functions in these settings remain to be explored.

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